

# Data modeling: CSCI E-106

Multiple Regression I

# Multiple Regression

- Multiple regression analysis is one of the most widely used of all statistical methods.
- a variety of multiple regression models
- basic statistical results in matrix form

# First-Order Model with Two Predictor Variables

- Two predictor variables:  $X_1; X_2$

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

- first-order model with two predictor variables: linear in the predictor variables.
  - $Y_i$  denotes as usual the response in the  $i$ th trial,
  - $X_{i1}$  and  $X_{i2}$  are the values of the two predictor variables in the  $i$ th trial.
  - Parameters of the model are  $\beta_0$ ,  $\beta_1$  and  $\beta_2$ , and the error term is  $\varepsilon_i$ .
- Assume  $E\{\varepsilon_i\} = 0 \Rightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
- called a regression surface

# Meaning of Regression Coefficients, cont'd

- $\beta_0$ : intercept in the regression plane;  $X_1 = 0, X_2 = 0$
- $\beta_1$ : the change in the mean response with  $\Delta X_1 = 1$  and  $X_2 = \text{constant}$ ;  $\frac{\partial E\{Y\}}{\partial X_1} = \beta_1$
- $\beta_2$ : the change in the mean response with  $X_1 = \text{constant}$  and  $\Delta X_2 = 1$ ;  $\frac{\partial E\{Y\}}{\partial X_2} = \beta_2$
- $\beta_1, \beta_2$ : **partial regression coefficients**  $\therefore$  they reflect the partial effect
- $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$ : additive or do not interact



# First-Order Model with More than Two Predictor Variables

- The first-order regression model with  $p - 1$  predictor variables:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i \\ &= \beta_0 + \sum_{k=1}^{p-1} \beta_k X_{ik} + \varepsilon_i \\ &= \sum_{k=0}^{p-1} \beta_k X_{ik} + \varepsilon_i, \quad \text{where } X_{i0} \equiv 1 \end{aligned}$$

- Assuming that  $E\{\varepsilon_i\} = 0$ , the response function (hyperplane) :

$$\Rightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{p-1} X_{p-1}$$

## First-Order Model with More than Two Predictor Variables, *cont'd*

- This response function is a hyperplane, which is a plane in more than two dimensions. It is no longer possible to picture this response surface
- $\beta_k$ : the change in the mean response  $E\{Y\}$  with  $\Delta X_k = 1$  and all other predictor variables are held constant
- additive and do not interact

# General linear Regression Model

- The general linear regression model with **normal error terms**:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, \dots, n \\ &= \beta_0 X_{i0} + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i \quad (X_{i0} \equiv 1) \\ &= \sum_{k=0}^{p-1} \beta_k X_{ik} + \varepsilon_i, \quad \text{where } X_{i0} \equiv 1 \end{aligned}$$

- Parameters:  $\beta_0, \beta_1, \dots, \beta_{p-1}$
- known constants:  $X_{i1}, X_{i2}, \dots, X_{ip-1}$
- $\varepsilon_i$ : independent  $N(0, \sigma^2)$

# General Linear Regression Model in Matrix Terms

The general linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, \dots, n$$

$$\begin{aligned} \underset{n \times 1}{\mathbf{Y}} &= \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} & \underset{n \times p}{\mathbf{X}} &= \begin{bmatrix} 1 & X_{11} & X_{12} & \cdots & X_{1,p-1} \\ 1 & X_{21} & X_{22} & \cdots & X_{2,p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & X_{n1} & X_{n2} & \cdots & X_{n,p-1} \end{bmatrix} \\ \underset{p \times 1}{\boldsymbol{\beta}} &= \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{p-1} \end{bmatrix} & \underset{n \times 1}{\boldsymbol{\varepsilon}} &= \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix} \end{aligned}$$

# General Linear Regression Model in Matrix Terms, cont'd

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$$\Rightarrow \underset{n \times 1}{Y} = \underset{n \times p}{X} \underset{p \times 1}{\beta} + \underset{n \times 1}{\varepsilon}$$

- $Y$ : vector of responses;
- $\beta$ : vector of parameters
- $X$ : matrix of constants
- $\varepsilon$ : vector of independent normal random variables;  
 $E\{\varepsilon\} = \mathbf{0}$ ;  $\underset{n \times n}{\sigma^2\{\varepsilon\}} = \sigma^2 I$
- Expectation and variance-covariance matrix of  $Y$ :

$$\underset{n \times 1}{E\{Y\}} = X\beta; \quad \underset{n \times 1}{\sigma^2\{Y\}} = \sigma^2 I$$

# Estimation of Regression Coefficients

- The general linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_{p-1} X_{i,p-1} + \varepsilon_i, \quad i = 1, \dots, n$$

- The **least squares criterion**: the values  $\beta_0, \dots, \beta_{p-1}$  minimize  $Q$

$$Q = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_{p-1} X_{i,p-1})^2$$

# Estimation of Regression Coefficients, cont'd

- The method of MLE leads to the same estimators of normal error regression model
- The likelihood function:

$$L(\boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} \exp \left[ -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \cdots - \beta_{p-1} X_{i,p-1})^2 \right]$$

Maximizing the likelihood function with respect to  $\beta_0, \beta_1, \dots, \beta_{p-1}$  leads to the estimators  $\mathbf{b}$

# Estimation of Regression Coefficients, cont'd

- the vector of the least squares estimated regression coefficients:
- The normal equations:

$$\underset{p \times 1}{\mathbf{b}} = \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{p-1} \end{bmatrix}$$
$$\mathbf{X}'\mathbf{X}\mathbf{b} = \mathbf{X}'\mathbf{Y}$$

- LSE:

$$\underset{p \times 1}{\mathbf{b}} = \underset{p \times p}{(\mathbf{X}'\mathbf{X})^{-1}} \underset{p \times 1}{\mathbf{X}'\mathbf{Y}}$$



# Design Matrix

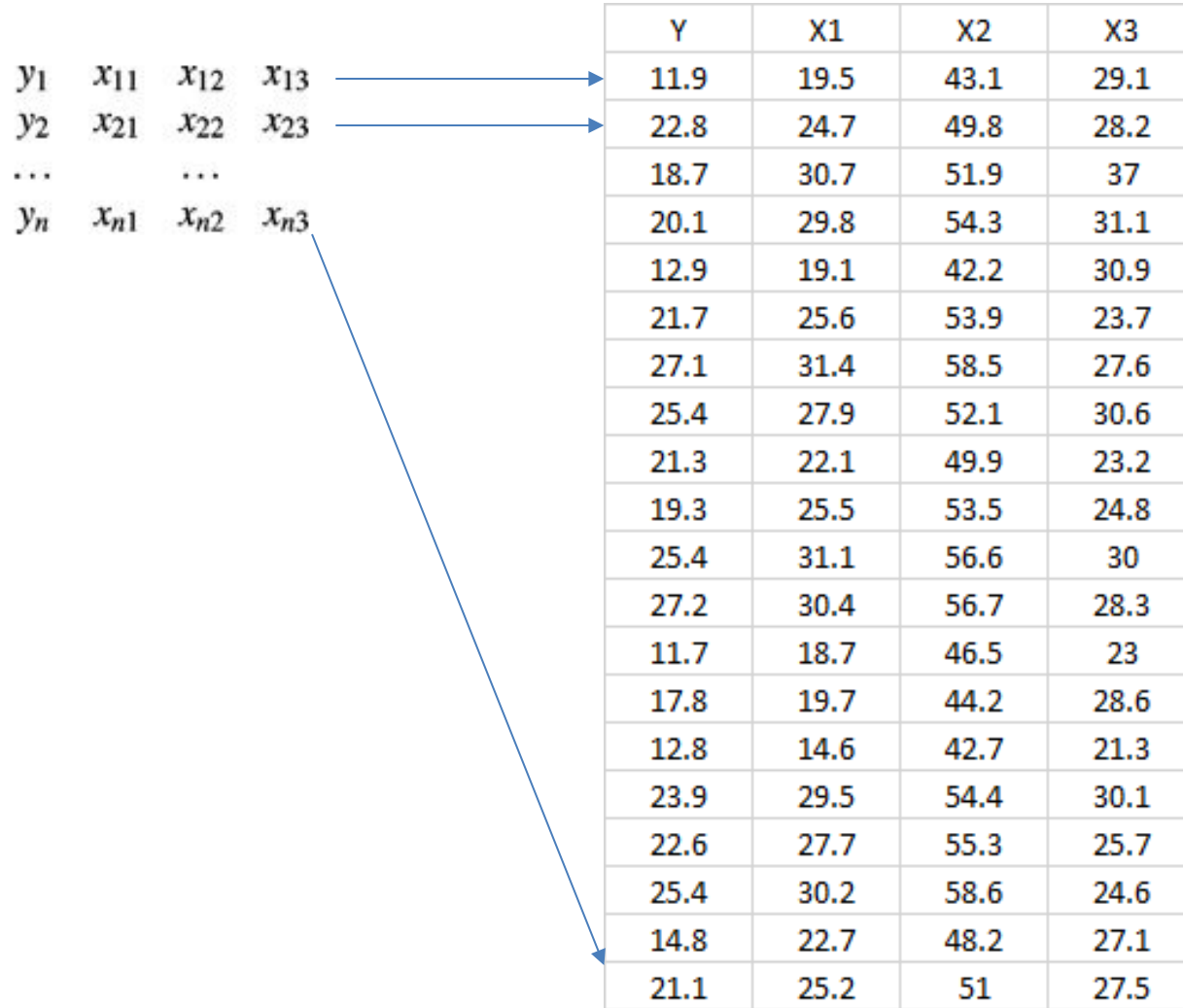
A **design matrix** is the numerical representation of your predictors used by regression models. It converts your dataset into a matrix where:

- Each **row** = one observation
- Each **column** = one predictor (including dummy variables and interaction terms)
- An **intercept column** of 1s is added unless you remove it explicitly

It's the object your model actually fits

# Design Matrix, *cont'd*

We start with some data with a response  $Y$  and three predictors,  $X_1$ ,  $X_2$ , and  $X_3$ . The data might be presented in tabular form like this



| $y$  | $x_1$ | $x_2$ | $x_3$ |
|------|-------|-------|-------|
| 11.9 | 19.5  | 43.1  | 29.1  |
| 22.8 | 24.7  | 49.8  | 28.2  |
| 18.7 | 30.7  | 51.9  | 37    |
| 20.1 | 29.8  | 54.3  | 31.1  |
| 12.9 | 19.1  | 42.2  | 30.9  |
| 21.7 | 25.6  | 53.9  | 23.7  |
| 27.1 | 31.4  | 58.5  | 27.6  |
| 25.4 | 27.9  | 52.1  | 30.6  |
| 21.3 | 22.1  | 49.9  | 23.2  |
| 19.3 | 25.5  | 53.5  | 24.8  |
| 25.4 | 31.1  | 56.6  | 30    |
| 27.2 | 30.4  | 56.7  | 28.3  |
| 11.7 | 18.7  | 46.5  | 23    |
| 17.8 | 19.7  | 44.2  | 28.6  |
| 12.8 | 14.6  | 42.7  | 21.3  |
| 23.9 | 29.5  | 54.4  | 30.1  |
| 22.6 | 27.7  | 55.3  | 25.7  |
| 25.4 | 30.2  | 58.6  | 24.6  |
| 14.8 | 22.7  | 48.2  | 27.1  |
| 21.1 | 25.2  | 51    | 27.5  |

# Design Matrix, *cont'd*

**intercept column of 1s is added**

|      | $\beta_0$ | $\beta_1$      | $\beta_2$      | $\beta_3$      |
|------|-----------|----------------|----------------|----------------|
| Y    | Intercept | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> |
| 11.9 | 1         | 19.5           | 43.1           | 29.1           |
| 22.8 | 1         | 24.7           | 49.8           | 28.2           |
| 18.7 | 1         | 30.7           | 51.9           | 37             |
| 20.1 | 1         | 29.8           | 54.3           | 31.1           |
| 12.9 | 1         | 19.1           | 42.2           | 30.9           |
| 21.7 | 1         | 25.6           | 53.9           | 23.7           |
| 27.1 | 1         | 31.4           | 58.5           | 27.6           |
| 25.4 | 1         | 27.9           | 52.1           | 30.6           |
| 21.3 | 1         | 22.1           | 49.9           | 23.2           |
| 19.3 | 1         | 25.5           | 53.5           | 24.8           |
| 25.4 | 1         | 31.1           | 56.6           | 30             |
| 27.2 | 1         | 30.4           | 56.7           | 28.3           |
| 11.7 | 1         | 18.7           | 46.5           | 23             |
| 17.8 | 1         | 19.7           | 44.2           | 28.6           |
| 12.8 | 1         | 14.6           | 42.7           | 21.3           |
| 23.9 | 1         | 29.5           | 54.4           | 30.1           |
| 22.6 | 1         | 27.7           | 55.3           | 25.7           |
| 25.4 | 1         | 30.2           | 58.6           | 24.6           |
| 14.8 | 1         | 22.7           | 48.2           | 27.1           |
| 21.1 | 1         | 25.2           | 51             | 27.5           |

$\beta$  vector

$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$

X=Design Matrix

Y vector

| Intercept | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> |
|-----------|----------------|----------------|----------------|
| 1         | 19.5           | 43.1           | 29.1           |
| 1         | 24.7           | 49.8           | 28.2           |
| 1         | 30.7           | 51.9           | 37             |
| 1         | 29.8           | 54.3           | 31.1           |
| 1         | 19.1           | 42.2           | 30.9           |
| 1         | 25.6           | 53.9           | 23.7           |
| 1         | 31.4           | 58.5           | 27.6           |
| 1         | 27.9           | 52.1           | 30.6           |
| 1         | 22.1           | 49.9           | 23.2           |
| 1         | 25.5           | 53.5           | 24.8           |
| 1         | 31.1           | 56.6           | 30             |
| 1         | 30.4           | 56.7           | 28.3           |
| 1         | 18.7           | 46.5           | 23             |
| 1         | 19.7           | 44.2           | 28.6           |
| 1         | 14.6           | 42.7           | 21.3           |
| 1         | 29.5           | 54.4           | 30.1           |
| 1         | 27.7           | 55.3           | 25.7           |
| 1         | 30.2           | 58.6           | 24.6           |
| 1         | 22.7           | 48.2           | 27.1           |
| 1         | 25.2           | 51             | 27.5           |

X =

| Y    |
|------|
| 11.9 |
| 22.8 |
| 18.7 |
| 20.1 |
| 12.9 |
| 21.7 |
| 27.1 |
| 25.4 |
| 21.3 |
| 19.3 |
| 25.4 |
| 27.2 |
| 11.7 |
| 17.8 |
| 12.8 |
| 23.9 |
| 22.6 |
| 25.4 |
| 14.8 |
| 21.1 |

Y =

# Design Matrix, *cont'd*

It will be more convenient to put this in a matrix/vector representation. The regression equation is then written as:

$$Y = X\beta + \varepsilon$$

Where  $Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$  and  $X = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \vdots & \dots & \dots & \dots \\ 1 & x_{n1} & x_{n2} & x_{n3} \end{bmatrix}$  and  $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$

$$Y = X\beta = \begin{bmatrix} Y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} \\ Y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} \\ Y_3 = \beta_0 + \beta_1 x_{31} + \beta_2 x_{32} + \beta_3 x_{33} \\ \vdots \\ Y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \beta_3 x_{n3} \end{bmatrix}$$

# Estimation of Regression Coefficients, cont'd

$$Y = X\beta$$

$$X'Y = X'X\beta$$

$$X'Y = (X'X)\beta$$

$$(X'X)^{-1}X'Y = (X'X)^{-1}(X'X)\beta$$

$$(X'X)^{-1}X'Y = \beta$$

Where  $X'$  is the transpose of the matrix and  $(X'X)^{-1}(X'X) = I$ , identity matrix. Therefore,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

# Idempotent Matrix

$$\hat{Y} = X\hat{\beta}$$

We plug in  $\hat{\beta} = (X'X)^{-1}X'Y$

$$\hat{Y} = X(X'X)^{-1}X'Y$$

Where  $H = X(X'X)^{-1}X'$  is called the hat matrix and idempotent matrix ( $H^2 = H$ )

$$H^2 = H * H$$
$$H^2 = X(X'X)^{-1} \underbrace{X'X(X'X)^{-1}}_{\text{Identity Matrix}} X'$$

$$H^2 = X(X'X)^{-1}X' = H$$

# Idempotent Matrix

$$\hat{Y} = HY$$

$$\hat{\varepsilon} = Y - \hat{Y}$$

$$\hat{\varepsilon} = Y - HY$$

$$\hat{\varepsilon} = (I - H)Y$$

$$\hat{\varepsilon}^2 = Y'(I - H)'(I - H)Y$$

$$\hat{\varepsilon}^2 = Y'(I - H)Y$$

Where  $(I - H)'(I - H) = (I - H)$  since H is idempotent matrix.

# General linear Regression Model, *cont'd*

- **p - 1 Predictor Variables:** When  $X_1, \dots, X_{p-1}$  represent p - 1 different predictor variables, there are no interaction effects between the predictor variables.
- **Qualitative Predictor Variables:** such as gender (male, female) or disability status(not disabled, partially disabled, fully disabled).
  - Indicator (dummy) variables are used to identify classes of a qualitative variable.
  - Example: Y is the length of hospital stay;  $X_1$  is age, and  $X_2$  is gender

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

$X_{i1}$ : patient's age

$$X_{i2}: \begin{cases} 1, & \text{where patient female} \\ 0, & \text{where patient male} \end{cases}$$



# Qualitative Predictor Variables

- The response function:  $E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
- The response function for male patients:  $X_2 = 0$   
 $E\{Y\} = \beta_0 + \beta_1 X_1$
- The response function for female patients:  $X_2 = 1$

$$E\{Y\} = (\beta_0 + \beta_2) + \beta_1 X_1$$

# Qualitative Predictors Variables, *cont'd*

- Field: (Qualitative variables) business, economics, the social, biological sciences
- Examples:
  - gender (M,F);
  - purchase status: purchase, no purchase
  - Disability status: not; partial; fully
- A study of innovation in the insurance industry
  - related the speed with which a particular insurance innovation is adopted (Y )
  - the size of the insurance firm ( $X_1$ )
  - the type of the firm ( $X_2$ )

# Qualitative Predictors Variables, *cont'd*

- Qualitative predictor with two classes:
  - indicator variable: take on 0 and 1
  - Ex: two indicator variables  $X_2, X_3$

$$X_2 = \begin{cases} 1 & \text{if stock company} \\ 0 & \text{otherwise} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{if mutual company} \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$$

# Qualitative Predictors Variables, *cont'd*

- intuitive approach of setting up an indicator variable: leads to computation difficulties
- Design matrix  $\mathbf{X}$  with  $n = 4$ :

$$\mathbf{X} = \begin{array}{c} \begin{array}{cc} & \begin{array}{ccc} X_1 & X_2 & X_3 \end{array} \\ \begin{bmatrix} 1 & X_{11} & 1 & 0 \\ 1 & X_{21} & 1 & 0 \\ 1 & X_{31} & 0 & 1 \\ 1 & X_{41} & 0 & 1 \end{bmatrix} \end{array} \\ \mathbf{X}'\mathbf{X} = \begin{bmatrix} 4 & \sum_{i=1}^4 X_{i1} & 2 & 2 \\ \sum_{i=1}^4 X_{i1} & \sum_{i=1}^4 X_{i1}^2 & \sum_{i=1}^2 X_{i1} & \sum_{i=3}^4 X_{i1} \\ 2 & \sum_{i=1}^2 X_{i1} & 2 & 0 \\ 2 & \sum_{i=3}^4 X_{i1} & 0 & 2 \end{bmatrix} \end{array}$$

- $\mathbf{X}$ : the first column = the sum of the last two columns  $\Rightarrow$  linearly dependent
- $\mathbf{X}'\mathbf{X}$ : no inverse

# Qualitative Predictors Variables, cont'd

Simple way out of the difficulty: drop one indicator variable

## Principle:

A qualitative variable with  $c$  classes will be represented by  $c - 1$  indicator variables, each taking on the values 0 and 1.

- Indicator variables: called dummy variables or binary variables

$$\begin{aligned}\text{Drop } X_3 &\Rightarrow Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i \\ &\Rightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2\end{aligned}$$

# Qualitative Predictors Variables, *cont'd*

Meaning of the regression coefficients:

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

- $X_2=0$

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2(0) = \beta_0 + \beta_1 X_1 \quad \text{Mutual Firms}$$

- $X_2=1$

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2(1) = (\beta_0 + \beta_2) + \beta_1 X_1 \quad \text{Stock Firms}$$

# Qualitative Predictors Variables, *cont'd*

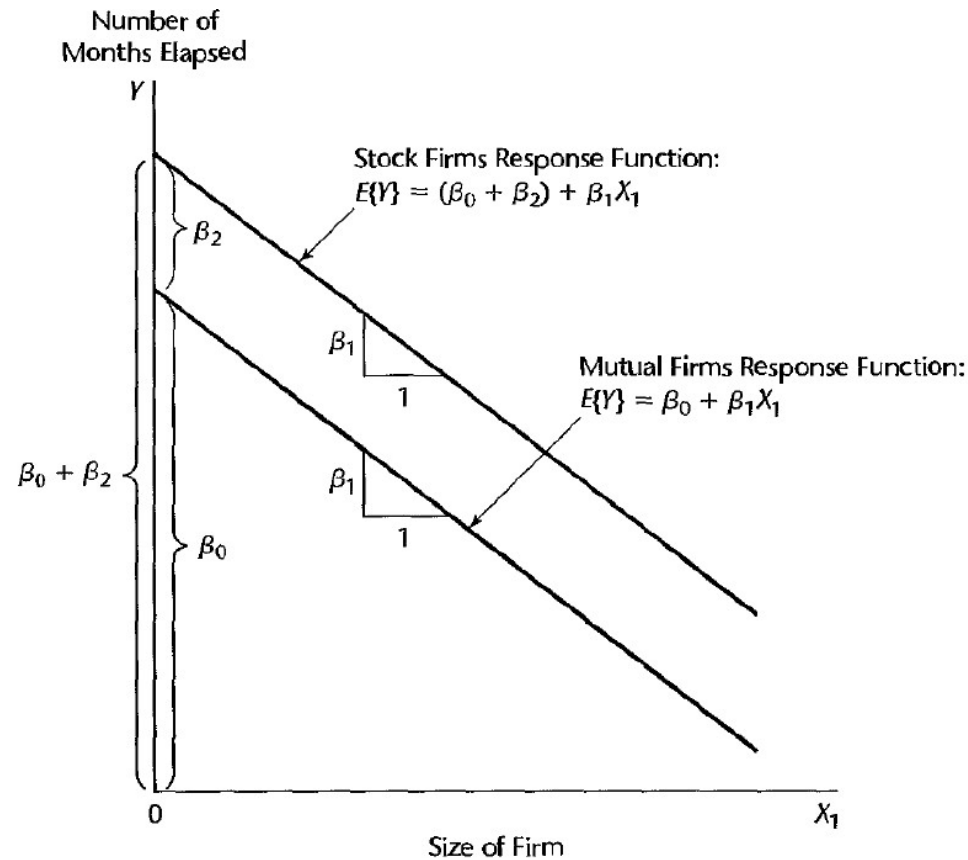


Figure : Illustration of Meaning of Regression Coefficient for Regression Model (8.33) with Indicator Variable  $X_2$ -Insurance Innovation Example.

# Qualitative Predictors Variables, *cont'd*

|      | (1)            | (2)               | (3)     | (4)       | (5)               |
|------|----------------|-------------------|---------|-----------|-------------------|
| Firm | Number of      | Size of Firm      | Type of | Indicator |                   |
| $i$  | Months Elapsed | (million dollars) | Firm    | Code      | $X_{i1}$ $X_{i2}$ |
|      | $Y_i$          | $X_{i1}$          |         | $X_{i2}$  |                   |
| 1    | 17             | 151               | Mutual  | 0         | 0                 |
| 2    | 26             | 92                | Mutual  | 0         | 0                 |
| 3    | 21             | 175               | Mutual  | 0         | 0                 |
| 4    | 30             | 31                | Mutual  | 0         | 0                 |
| 5    | 22             | 104               | Mutual  | 0         | 0                 |
| 6    | 0              | 277               | Mutual  | 0         | 0                 |
| 7    | 12             | 210               | Mutual  | 0         | 0                 |
| 8    | 19             | 120               | Mutual  | 0         | 0                 |
| 9    | 4              | 290               | Mutual  | 0         | 0                 |
| 10   | 16             | 238               | Mutual  | 0         | 0                 |
| 11   | 28             | 164               | Stock   | 1         | 164               |
| 12   | 15             | 272               | Stock   | 1         | 272               |
| 13   | 11             | 295               | Stock   | 1         | 295               |
| 14   | 38             | 68                | Stock   | 1         | 68                |
| 15   | 31             | 85                | Stock   | 1         | 85                |
| 16   | 21             | 224               | Stock   | 1         | 224               |
| 17   | 20             | 166               | Stock   | 1         | 166               |
| 18   | 13             | 305               | Stock   | 1         | 305               |
| 19   | 30             | 124               | Stock   | 1         | 124               |
| 20   | 14             | 246               | Stock   | 1         | 246               |



# Qualitative Predictors Variables, *cont'd*

- studied 10 mutual firms and 10 stock firms
- The fitted of regression model:

$$\hat{Y} = 33.87407 - 0.10174X_1 + 8.05547X_2$$

- Interested in the effect of type of firm ( $X_2$ )
- a 95% confidence interval for  $\beta_2$ :

$$4.98 \leq \beta_2 \leq 11.13 \quad (t(0.975; 17) = 2.110)$$

- Test:

$$H_0 : \beta_2 = 0 \quad \text{vs.} \quad H_a : \beta_2 \neq 0$$

$\Rightarrow$  lead to  $H_a$ : type of firm has an effect ( $\alpha = 0.05$ )

# Qualitative Predictors Variables, *cont'd*

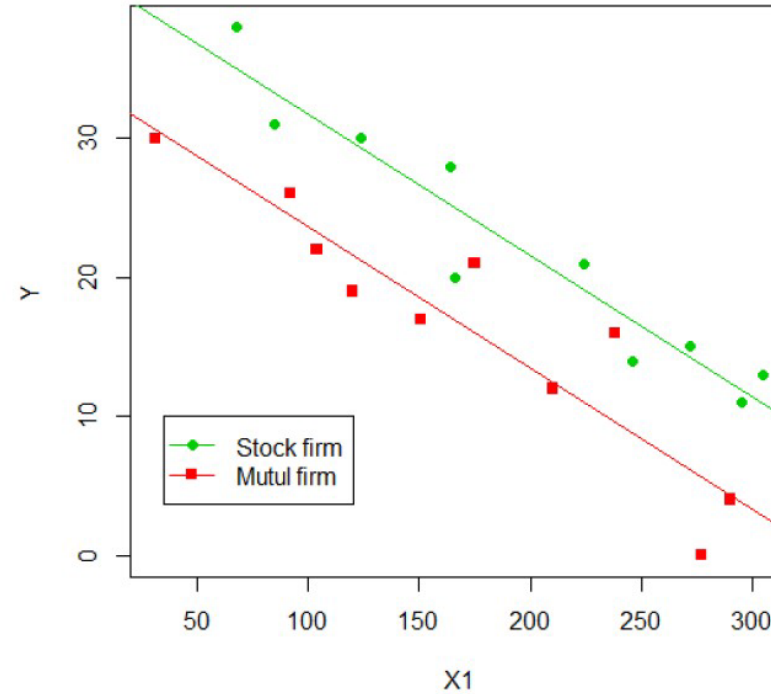
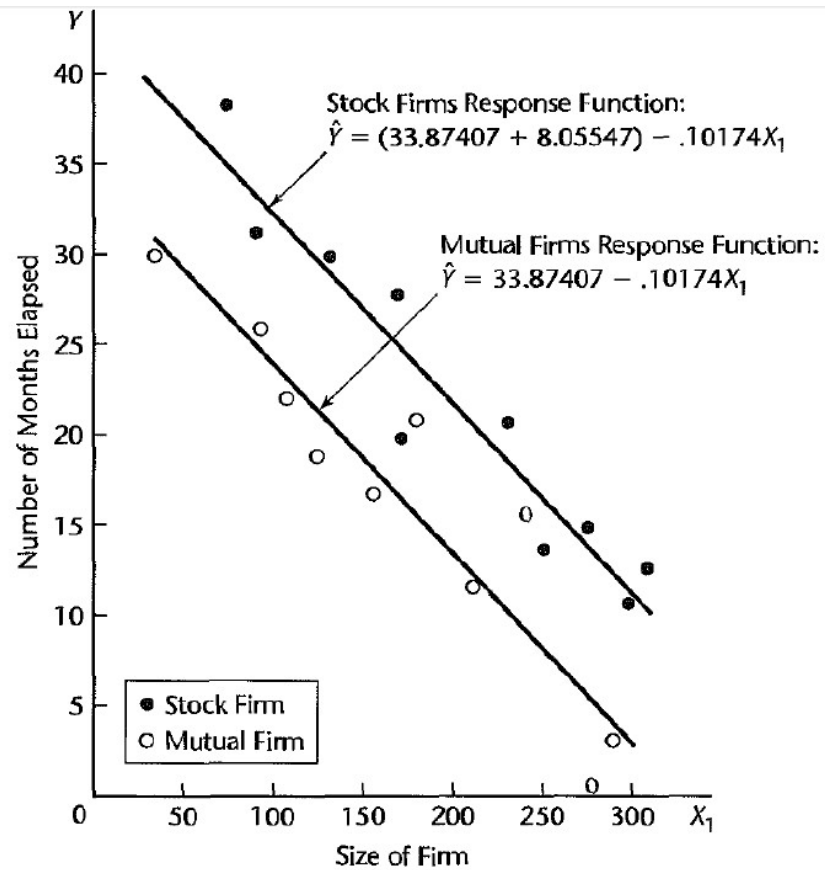
**Figure :** Regression Results for Fit of Regression Model (8.33)-Insurance Innovation Example.

| (a) Regression Coefficients |                                  |                              |        |
|-----------------------------|----------------------------------|------------------------------|--------|
| Regression Coefficient      | Estimated Regression Coefficient | Estimated Standard Deviation | t*     |
| $\beta_0$                   | 33.87407                         | 1.81386                      | 18.68  |
| $\beta_1$                   | -.10174                          | .00889                       | -11.44 |
| $\beta_2$                   | 8.05547                          | 1.45911                      | 5.52   |

| (b) Analysis of Variance |          |    |        |
|--------------------------|----------|----|--------|
| Source of Variation      | SS       | df | MS     |
| Regression               | 1,504.41 | 2  | 752.20 |
| Error                    | 176.39   | 17 | 10.38  |
| Total                    | 1,680.80 | 19 |        |

# Qualitative Predictors Variables, *cont'd*



**Figure :** Fitted Regression Function for Regression Model (8.33) Indicator Innovation Example.

# Qualitative Predictors Variables, *cont'd*

Why did we not simply fit separate regressions for stock firms and mutual firms

1. The model assumes equal slopes and the same constant error term variance for each type of firm, the common slope  $\beta_1$  can best be estimated by pooling the two types of firms
2. Other inferences,  $\beta_0$ ,  $\beta_2$ , can be made more precisely by working with one regression model containing an indicator variable since more *df* associated with *MSE*.

# More than two classes

- The regression of tool wear ( $Y$ ) on tool speed  $X_1$  and tool Model (qualitative: M1, M2, M3, M4)
- Require three indicator variables:

$$X_2 = \begin{cases} 1 & \text{if tool model M1} \\ 0 & \text{otherwise} \end{cases} \quad X_3 = \begin{cases} 1 & \text{if tool model M2} \\ 0 & \text{otherwise} \end{cases}$$
$$X_4 = \begin{cases} 1 & \text{if tool model M3} \\ 0 & \text{otherwise} \end{cases}$$

# More than two classes, cont'd

## First-order model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \varepsilon_i$$

$$\Rightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

$$\Leftrightarrow \text{Tool model M4: } E\{Y\} = \beta_0 + \beta_1 X_1$$

$$\text{Tool model M1: } E\{Y\} = (\beta_0 + \beta_2) + \beta_1 X_1$$

$$\text{Tool model M2: } E\{Y\} = (\beta_0 + \beta_3) + \beta_1 X_1$$

$$\text{Tool model M3: } E\{Y\} = (\beta_0 + \beta_4) + \beta_1 X_1$$

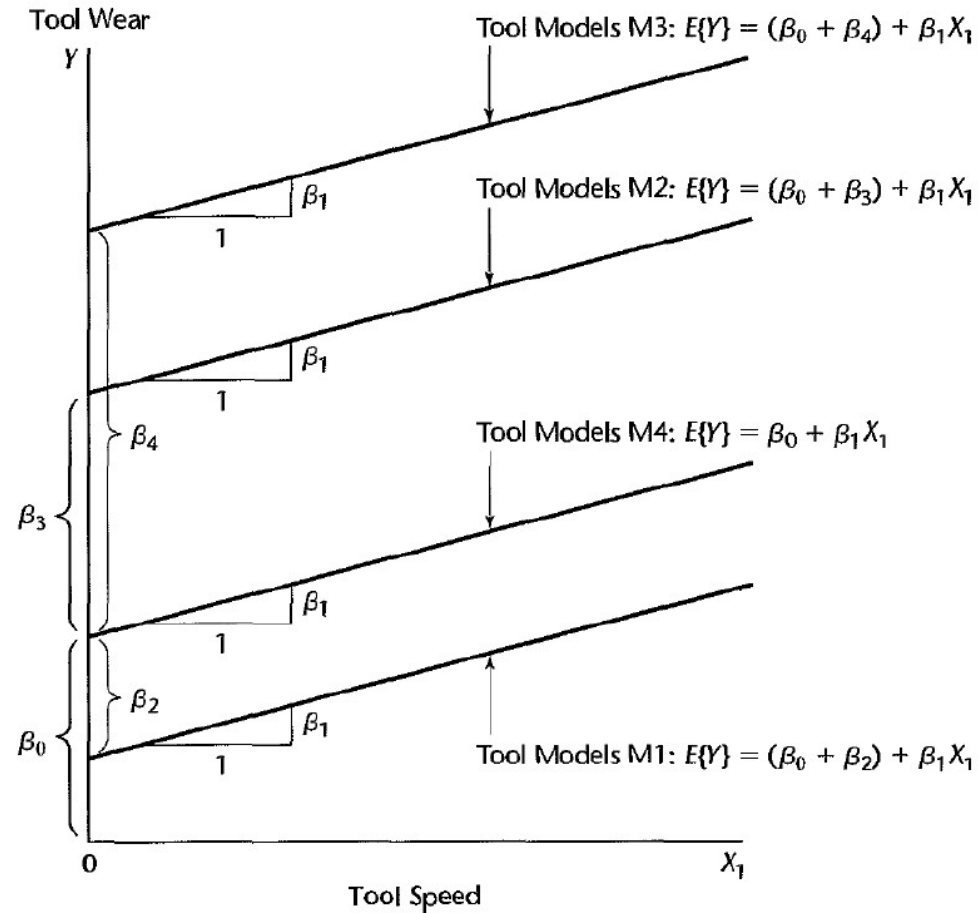
| Tool Model | $X_1$    | $X_2$ | $X_3$ | $X_4$ |
|------------|----------|-------|-------|-------|
| M1         | $X_{i1}$ | 1     | 0     | 0     |
| M2         | $X_{i1}$ | 0     | 1     | 0     |
| M3         | $X_{i1}$ | 0     | 0     | 1     |
| M4         | $X_{i1}$ | 0     | 0     | 0     |

# More than two classes, cont'd

- the regression of tool wear on tool speed is **linear**, with **the same slope** for all four tool models
- $\beta_2, \beta_3, \beta_4$ : how much higher (lower) the response functions for tool models (M1,M2,M3) are than the one for M4
- always compared with the class for which  $X_2 = X_3 = X_4 = 0$

# More than two classes, cont'd

Figure : Illustration of Regression Model (8.36) - Tool Wear Example.





# Time Series Applications

- Economists; business analysis
- Using indicator variables:
  - year: peacetime; wartime

$$Y_t = \beta_0 + \beta_1 X_{t1} + \beta_2 X_{t2} + \varepsilon_t, \quad t = 1, \dots, n$$

where  $X_{t1} = \text{income}$

$$X_{t2} = \begin{cases} 1 & \text{if period } t \text{ peace time} \\ 0 & \text{otherwise} \end{cases}$$

- monthly or quarterly: seasonal effect
- Time series data: susceptible to correlated error terms (Chap. 12)

# Some Considerations in Using Indicator Variables

Allocated codes: arbitrary; other numbers;

- define a metric for the classes of the qualitative variable

| Class           | $X_1$ | $E\{Y\}$                                             |
|-----------------|-------|------------------------------------------------------|
| Frequent user   | 3     | $E\{Y\} = \beta_0 + \beta_1(3) = \beta_0 + 3\beta_1$ |
| Occasional user | 2     | $E\{Y\} = \beta_0 + \beta_1(2) = \beta_0 + 2\beta_1$ |
| Nonuser         | 1     | $E\{Y\} = \beta_0 + \beta_1(1) = \beta_0 + \beta_1$  |

$$Y_i = \beta_0 + \beta_1 X_{i1} + \varepsilon_i$$

The key implication:

$$\begin{aligned} & E\{Y|frequent\ user\} - E\{Y|ocasional\ user\} \\ &= E\{Y|ocasional\ user\} - E\{Y|nonuser\} = \beta_1 \end{aligned}$$

# Some Considerations in Using Indicator Variables, cont'd

Indicator variables: make no assumptions about the spacing of the classes;

| Class           | $X_1$ | $X_2$ |
|-----------------|-------|-------|
| Frequent user   | 1     | 0     |
| Occasional user | 0     | 1     |
| Nonuser         | 0     | 0     |

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

$$\beta_1: E\{Y|\text{frequent user}\} - E\{Y|\text{nonuser}\}$$

$$\beta_2: E\{Y|\text{ocasional user}\} - E\{Y|\text{nonuser}\}$$

- $\beta_1 - \beta_2$ : the differential effect between frequent user and occasional user
- if  $\beta_1 = 2\beta_2 \Rightarrow$  equal spacing between the three classes

# Some Considerations in Using Indicator Variables, cont'd

- Indicator variables can be used even if the predictor variable is **quantitative**.
  - transformed by grouping into classes
  - age: 21, 21-34, 35-49
- **information** about the original variable may be thrown away
- **additional parameters** into the model: reducing the df associated with *MSE*

# Some Considerations in Using Indicator Variables, cont'd

## Other codings:

- First coding:

$$X_{i2} = \begin{cases} 1 & \text{if stock company} \\ -1 & \text{if mutual company} \end{cases}$$

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$

$$E\{Y\} = (\beta_0 + \beta_2) + \beta_1 X_1 \quad \text{Stock firms}$$

$$E\{Y\} = (\beta_0 - \beta_2) + \beta_1 X_1 \quad \text{Mutual firms}$$

$\Rightarrow \beta_0$ : "average" intercept of the regression line

- two regression lines are the same:

$$H_0 : \beta_2 = 0 \text{ vs. } H_a : \beta_2 \neq 0$$

# Some Considerations in Using Indicator Variables, cont'd

- Second coding:
  - using **indicator variables** for **each of the  $c$  classes** of the qualitative variable
  - **drop the intercept term**

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i$$

where:  $X_{i1}$  = size of firm;  $X_{i2} = \begin{cases} 1 & \text{if stock company} \\ 0 & \text{otherwise} \end{cases}$

$$X_{i3} = \begin{cases} 1 & \text{if mutual company} \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow E\{Y\} = \beta_2 + \beta_1 X_1 \quad \text{Stock firms}$$

$$E\{Y\} = \beta_3 + \beta_1 X_1 \quad \text{Mutual firms}$$

- The same regression line:  $H_0 : \beta_2 = \beta_3$  vs.  $H_a : \beta_2 \neq \beta_3$



# Modeling Interactions between Quantitative and Qualitative Predictors

- the possibility of interaction effects: the size of firm and type of firm

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i$$

where  $X_{i1}$  = *size of firm*

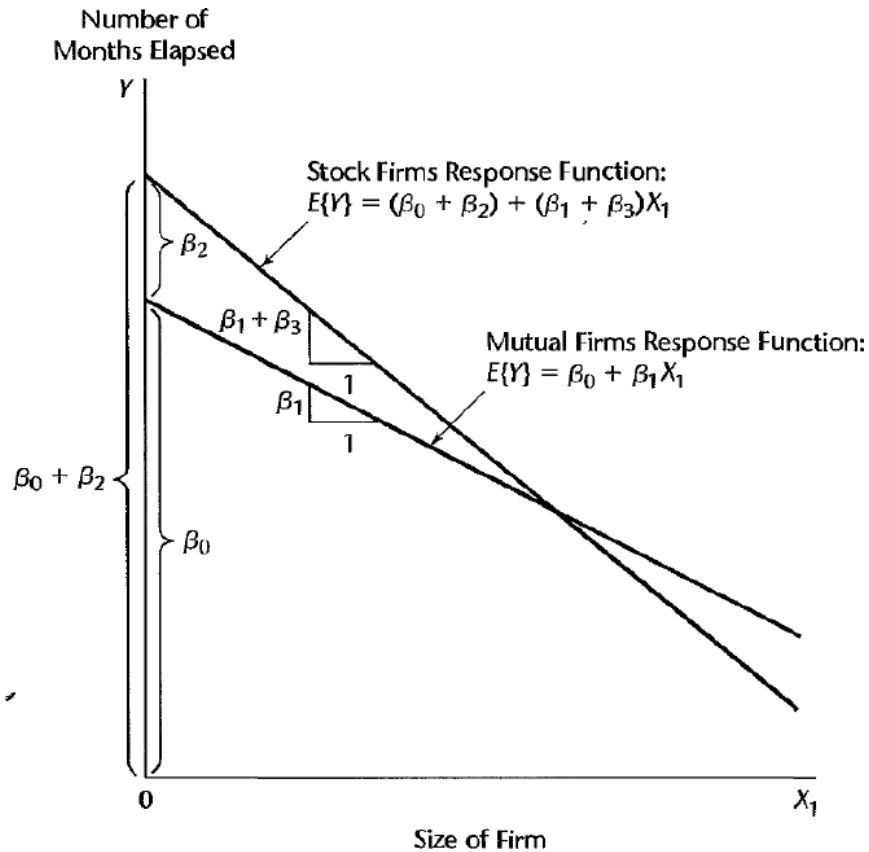
$$X_{i2} = \begin{cases} 1 & \text{if stock company} \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$$

$$\Leftrightarrow E\{Y\} = \beta_0 + \beta_1 X_1 + \beta_2(0) + \beta_3(0) = \beta_0 + \beta_1 X_1 \text{ Mutual firms}$$

$$E\{Y\} = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X_1 \text{ Stock firms}$$

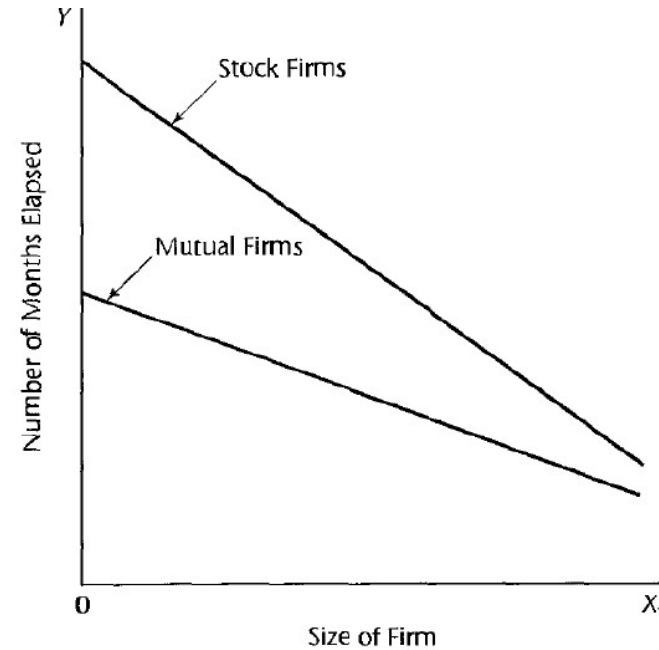
## Modeling Interactions between Quantitative and Qualitative Predictors, cont'd



- Both the **intercept** and the **slope** differ for the two classes in the response function
- The effect of the qualitative predictor variable can be studied only by comparing the regression functions within the scope of the model for the different classes of the qualitative variable



# Modeling Interactions between Quantitative and Qualitative Predictors, cont'd



**Figure :** Another Illustration of Regression Model (8.49) with Indicator Variable  $X_2$  and Interaction Term- Insurance Innovation Example.

# Modeling Interactions between Quantitative and Qualitative Predictors, cont'd

## Example: Insurance Innovation

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i$$

Figure : Regression Result for Fit of Regression Model (8.49) with Interaction Term - Insurance Innovation Example.

| (a) Regression Coefficients |                                  |                              |       |
|-----------------------------|----------------------------------|------------------------------|-------|
| Regression Coefficient      | Estimated Regression Coefficient | Estimated Standard Deviation | t*    |
| $\beta_0$                   | 33.83837                         | 2.44065                      | 13.86 |
| $\beta_1$                   | -.10153                          | .01305                       | -7.78 |
| $\beta_2$                   | 8.13125                          | 3.65405                      | 2.23  |
| $\beta_3$                   | -.0004171                        | .01833                       | -.02  |

| (b) Analysis of Variance |          |    |        |
|--------------------------|----------|----|--------|
| Source of Variation      | SS       | df | MS     |
| Regression               | 1,504.42 | 3  | 501.47 |
| Error                    | 176.38   | 16 | 11.02  |
| Total                    | 1,680.80 | 19 |        |

# Modeling Interactions between Quantitative and Qualitative Predictors, cont'd

## Example: Insurance Innovation

$$H_0 : \beta_3 = 0 \text{ vs. } H_a : \beta_3 \neq 0 \Rightarrow |t^*| = 0.02 < t(0.975; 16)$$

⇒ no interaction effects

```
> summary(lm(Y~X1+X2+X1*X2))
Call:
lm(formula = Y ~ X1 + X2 + X1 * X2)

Residuals:
    Min       1Q   Median       3Q      Max
-5.7144 -1.7064 -0.4557  1.9311  6.3259

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 33.8383695   2.4406498   13.864 2.47e-10 ***
X1          -0.1015306   0.0130525   -7.779 7.97e-07 ***
X2           8.1312501   3.6540517    2.225  0.0408 *
X1:X2       -0.0004171   0.0183312   -0.023  0.9821
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

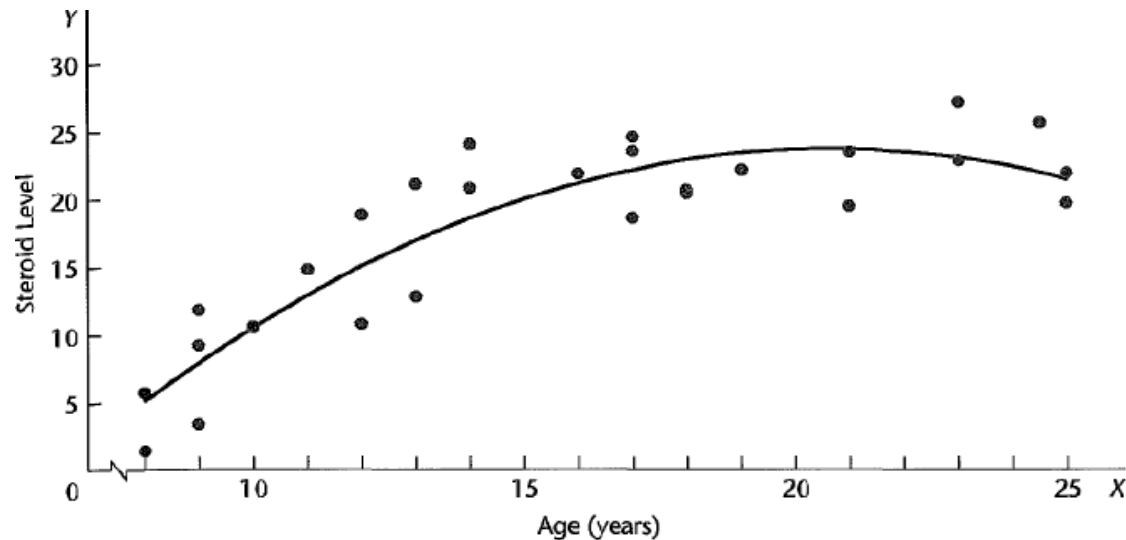
Residual standard error: 3.32 on 16 degrees of freedom
Multiple R-squared:  0.8951,    Adjusted R-squared:  0.8754
F-statistic: 45.49 on 3 and 16 DF,  p-value: 4.675e-08
```

# Polynomial Regression

- Special cases of the general regression model
- Contain squared and higher order terms of the predicted variables makes the response function curvilinear.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

$$\text{let } X_i = X_{i1} \text{ and } X_i^2 = X_{i2} \Rightarrow Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$$



# Transformed Variables

- complex, curvilinear response functions, yet still are special cases of the general linear regression model.
- Examples:

$$\log Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i \quad (Y'_i = \log Y_i)$$

$$Y_i = \frac{1}{\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i} \quad (Y'_i = \frac{1}{Y_i})$$

# Interaction Effects and Combination of Cases

An example of a nonadditive regression model with  $X_1, X_2$ : (complex)

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} X_{i2} + \varepsilon_i \\ &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \varepsilon_i, \quad (X_{i3} = X_{i1} X_{i2}) \end{aligned}$$

By cross-product term:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2}^2 + \beta_3 X_{i2} + \beta_4 X_{i2}^2 + \beta_5 X_{i1} X_{i2} + \varepsilon_i \\ &= \beta_0 + \beta_1 Z_{i1} + \beta_2 Z_{i2} + \beta_3 Z_{i3} + \beta_4 Z_{i4} + \beta_5 Z_{i5} + \varepsilon_i \end{aligned}$$

# Meaning of Linear in General Linear Regression Model

- A regression model is **linear in the parameters**:

$$Y_i = c_{i0}\beta_0 + c_{i1}\beta_1 + c_{i2}\beta_2 + \cdots + c_{i,p-1}\beta_{p-1} + \varepsilon_i$$

where  $c_{ik}$ ,  $k = 0, \dots, p - 1$  are coefficients involving the predictor variables

- Illustration: **nonlinear**

$$Y_i = \beta_0 \exp(\beta_1 X_i) + \varepsilon_i$$

# Fitted Values and Residuals

- The vector of  $\hat{Y}_i$  and the vector of  $e_i = Y_i - \hat{Y}_i$ :

$$\hat{\mathbf{Y}}_{n \times 1} = \begin{bmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \vdots \\ \hat{Y}_n \end{bmatrix} = \mathbf{X}\mathbf{b} = \mathbf{H}\mathbf{Y} \quad (\mathbf{H}_{n \times n} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')$$

$$\mathbf{e}_{n \times 1} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\mathbf{b} = (\mathbf{I} - \mathbf{H})\mathbf{Y} \quad (= (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon})$$



## Fitted Values and Residuals, cont'd

the variance-covariance matrix of the residuals:

$$\sigma^2_{n \times n} \{\mathbf{e}\} = \sigma^2 (\mathbf{I} - \mathbf{H})$$

$$\Rightarrow \text{(estimated by)} \quad s^2_{n \times n} \{\mathbf{e}\} = MSE (\mathbf{I} - \mathbf{H})$$

# Analysis of Variance Results, cont'd

| Source     | df    | SS  | MS                            | F                     |
|------------|-------|-----|-------------------------------|-----------------------|
| Regression | p     | SSR | $MSR = \frac{SSR}{p}$         | $F = \frac{MSR}{MSE}$ |
| Residual   | N-p-1 | SSE | $MSE = \frac{SSE}{N - p - 1}$ |                       |
| Total      | N-1   | SST |                               |                       |

Where:

N = sample size

p = number of predictors (not counting the intercept)

# Analysis of Variance Results, cont'd

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$  SSR and df will be displayed at granular level for each independent variable in R output (where  $p=3$ )

| Source     | df    |   | SS  |                     | MS                            |                                          | F                          |
|------------|-------|---|-----|---------------------|-------------------------------|------------------------------------------|----------------------------|
| Regression | 3     | 1 | SSR | $SSR(X_1)$          | MSR                           | $MSR(X_1) = \frac{SSR(X_1)}{1}$          | $F = \frac{MSR(X_1)}{MSE}$ |
|            |       | 1 |     | $SSR(X_2/X_1)$      |                               | $MSR(X_2) = \frac{SSR(X_2/X_1)}{1}$      | $F = \frac{MSR(X_2)}{MSE}$ |
|            |       | 1 |     | $SSR(X_3/X_1, X_2)$ |                               | $MSR(X_3) = \frac{SSR(X_3/X_1, X_2)}{1}$ | $F = \frac{MSR(X_3)}{MSE}$ |
| Residual   | N-p-1 |   | SSE |                     | $MSE = \frac{SSE}{N - p - 1}$ |                                          |                            |
| Total      | N-1   |   | SST |                     |                               |                                          |                            |

Where:

N = sample size

p = number of predictors (not counting the intercept)

$$SSR = SSR(X_1) + SSR(X_2/X_1) + SSR(X_3/X_1, X_2)$$

# Analysis of Variance Results, cont'd

## Notations:

- Assume  $X_1$  is in the model
  - $SSR(X_1)$ : The regression sum of squares
  - $SSE(X_1)$ : The error sum of squares
  - measure the marginal effect of adding  $X_2$  (another variable) to the regression model when  $X_1$  is already in the model
    - $SSR(X_2|X_1)$ : The extra sum of squares gained by adding  $X_2$
- Assume  $X_1$  and  $X_2$  are in the model
  - $SSR(X_1, X_2)$ : The regression sum of squares
  - $SSE(X_1, X_2)$ : The error sum of squares
  - measure the marginal effect of adding  $X_3$  (another variable) to the regression model when  $X_1$  and  $X_2$  are already in the model
    - $SSR(X_3|X_1, X_2)$ : The extra sum of squares gained by adding  $X_3$

# F Test for Regression Relation

- Test whether there is a regression relation between  $Y$  and  $X$ :

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0$$

$$H_a : \text{not all } \beta_k \text{ } (k = 1, \dots, p - 1) \text{ equal zero}$$

$$\Rightarrow \text{test statistic: } F^* = \frac{MSR}{MSE}$$

- The **decision rule** to control the Type I error at  $\alpha$ :

If  $F^* \leq F(1 - \alpha; p - 1, n - p)$ , conclude  $H_0$

If  $F^* > F(1 - \alpha; p - 1, n - p)$ , conclude  $H_a$

# Coefficient of Multiple Determination

- The coefficient of multiple determination:  $R^2$

$$R^2 = \frac{SSR}{SSTO} = 1 - \frac{SSE}{SSTO}$$

- Measures the proportionate reduction of total variation in  $Y$  associated with  $X_1, \dots, X_{p-1}$
- $0 \leq R^2 \leq 1$
- Adding more  $X$  variable  $\Rightarrow R^2 \uparrow$ 
  - $SSE$  can never become larger with more  $X$  variables
  - $SSTO$  is always the same of a given set of responses

# Coefficient of Multiple Determination, cont'd

- The adjusted coefficient of multiple determination:  $R_a^2$

$$R_a^2 = 1 - \frac{\frac{SSE}{n-p}}{\frac{SSTO}{n-1}} = 1 - \left( \frac{n-1}{n-p} \right) \frac{SSE}{SSTO}$$

- $R_a^2$  becomes smaller when another  $X$  is introduced into the model  
( $\because SSE \downarrow$ )
- Coefficient of Multiple Correlation:  $R$

$$R = \sqrt{R^2}$$

# Inferences about Regression Parameters

Unbiased:  $E\{\mathbf{b}\} = \boldsymbol{\beta}$

The variance-covariance matrix  $\sigma^2\{\mathbf{b}\}$ :

$$\begin{aligned}\sigma^2\{\mathbf{b}\}_{p \times p} &= \begin{bmatrix} \sigma^2\{b_0\} & \sigma\{b_0, b_1\} & \cdots & \sigma\{b_0, b_{p-1}\} \\ \sigma\{b_1, b_0\} & \sigma^2\{b_1\} & \cdots & \sigma\{b_1, b_{p-1}\} \\ \vdots & \vdots & & \vdots \\ \sigma\{b_{p-1}, b_0\} & \sigma\{b_{p-1}, b_1\} & \cdots & \sigma^2\{b_{p-1}\} \end{bmatrix} \\ &= \sigma^2\{\mathbf{b}\}_{p \times p} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$



# Inferences about Regression Parameters, cont'd

The estimated variance-covariance matrix  $s^2\{\mathbf{b}\}$ :

$$s^2\{\mathbf{b}\}_{p \times p} = \begin{bmatrix} s^2\{b_0\} & s\{b_0, b_1\} & \cdots & s\{b_0, b_{p-1}\} \\ s\{b_1, b_0\} & s^2\{b_1\} & \cdots & s\{b_1, b_{p-1}\} \\ \vdots & \vdots & & \vdots \\ s\{b_{p-1}, b_0\} & s\{b_{p-1}, b_1\} & \cdots & s^2\{b_{p-1}\} \end{bmatrix}$$
$$= MSE(\mathbf{X}'\mathbf{X})^{-1}$$

# Interval Estimation of $\beta_k$

- For the normal error regression model:

$$\frac{b_k - \beta_k}{s\{b_k\}} \sim t(n - p) \quad k = 0, 1, \dots, p - 1$$

- The confidence limits for  $\beta_k$  with  $1 - \alpha$  confidence coefficient:

$$b_k \pm t(1 - \alpha/2; n - p)s\{b_K\}$$

# Interval Estimation of $\beta_k$ , cont'd

- Tests for  $\beta_k$ :

$$H_0 : \beta_k = 0 \quad H_a : \beta_k \neq 0$$

$$\Rightarrow t^* = \frac{b_k}{s\{b_k\}}$$

$\Rightarrow$  The decision rule:

If  $|t^*| \leq t(1 - \alpha/2; n - p)$ , conclude  $H_0$

Otherwise conclude  $H_a$

# Interval Estimation of $E\{Y_h\}$

- Given values of  $X_1, \dots, X_{p-1}$ :  $X_{h,1}, \dots, X_{h,p-1}$

$$\underset{p \times 1}{\mathbf{X}_h} = \begin{bmatrix} 1 \\ X_{h1} \\ \vdots \\ X_{h,p-1} \end{bmatrix}$$

- The mean response  $E\{Y_h\}$ :

$$E\{Y_h\} = \mathbf{X}_h' \boldsymbol{\beta}$$

# Interval Estimation of $E\{Y_h\}$ , cont'd

- The estimated mean response  $\hat{Y}_h$ :

$$\begin{aligned}\hat{Y}_h &= \mathbf{X}_h' \mathbf{b} \\ \Rightarrow E\{\hat{Y}_h\} &= \mathbf{X}_h' \boldsymbol{\beta} = E\{Y_h\} \quad (\textit{Unbiased}) \\ \sigma^2\{\hat{Y}_h\} &= \sigma^2 \mathbf{X}_h' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}_h = \sigma^2\{\hat{Y}_h\} = \mathbf{X}_h' \boldsymbol{\sigma}^2\{\mathbf{b}\} \mathbf{X}_h\end{aligned}$$

- The estimated variance  $s^2\{\hat{Y}_h\}$ :

$$s^2\{\hat{Y}_h\} = \text{MSE}(\mathbf{X}_h' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}_h) = \mathbf{X}_h' \mathbf{s}^2\{\mathbf{b}\} \mathbf{X}_h$$

- The  $1 - \alpha$  confidence limits for  $E\{Y_h\}$ :

$$\hat{Y}_h \pm t(1 - \alpha/2; n - p) s\{\hat{Y}_h\}$$

# Confidence region for regression surface

- The Working-Hotelling confidence band for the regression line
- Boundary points of the confidence region at  $X_h$ :

$$\hat{Y}_h \pm Ws\{\hat{Y}_h\}$$

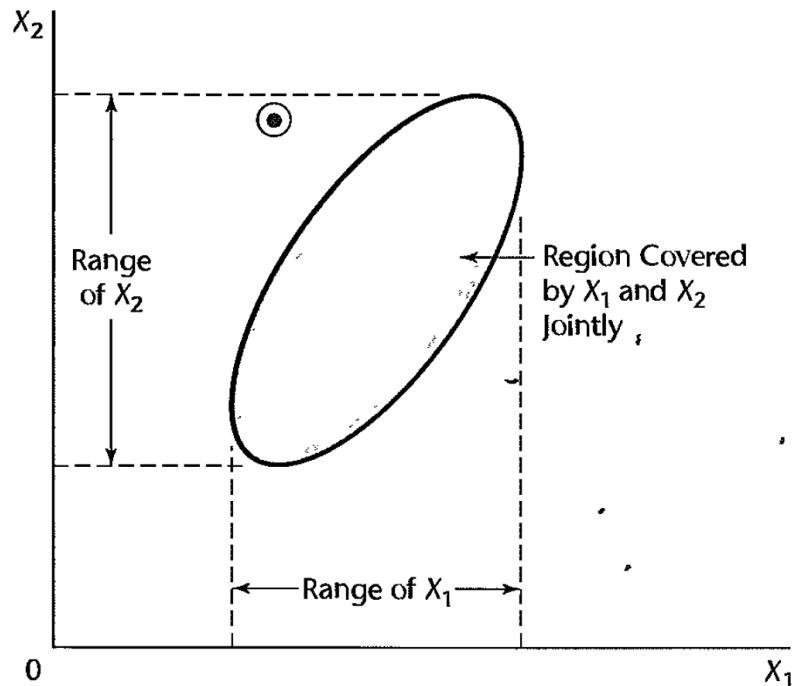
$$W^2 = pF(1 - \alpha; p, n - p)$$

# Prediction of New Observation $Y_{h(\text{new})}$

- The  $1 - \alpha$  prediction limits for a new observation  $Y_{h(\text{new})}$  at  $X_h$ :
  - $\hat{Y}_h \pm t(1 - \frac{\alpha}{2}; n - p) s\{pred\}$
  - where  $s^2\{pred\} = \text{MSE} + s^2\{\hat{Y}_h\} = \text{MSE} (1 + X_h' (X'X)^{-1} X_h)$
- Prediction of mean of  $m$  new observations at  $X_h$ :
  - $\hat{Y}_h \pm t(1 - \frac{\alpha}{2}; n - p) s\{predmean\}$
  - Where  $s^2\{predmean\} = \frac{\text{MSE}}{m} + s^2\{\hat{Y}_h\} = \text{MSE} (\frac{1}{m} + X_h' (X'X)^{-1} X_h)$

# Caution about Hidden Extrapolations

- Danger: the model may not be appropriate when it is **extended outside the region of the observations**.
- In multiple regression, it is particularly easy to lose track of this region since the levels of  $X_1, \dots, X_{p-1}$  jointly define the region. Thus, one cannot merely look at the ranges of each predictor variable.

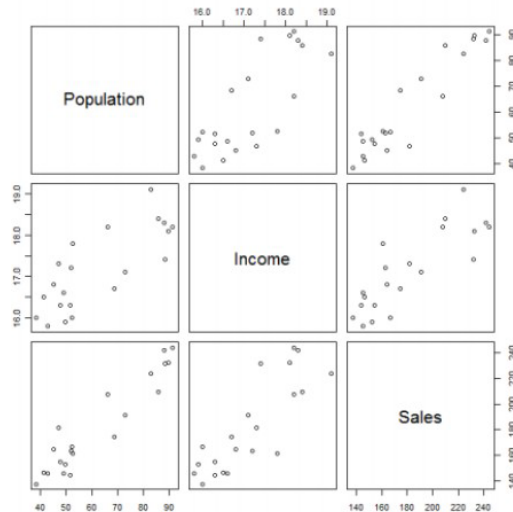


- The circled dot is within the ranges of the predictor variables  $X_1$  and  $X_2$  individually, yet is well outside the joint region of observations.
- It is easy to spot this extrapolation when there are only two predictor variables, but it becomes much more difficult when the number of predictor variables is large.



# Diagnostics and Remedial Measures

- Diagnostics play an important role in the development and evaluation of multiple regression models.
- Most of the diagnostic procedures for simple linear regression that we described in previous lectures carry over directly to multiple regression.
- Some important ones will be discussed later.



(b) Correlation Matrix

|           | SALES | TARGETPOP | DISPOINC |
|-----------|-------|-----------|----------|
| SALES     | 1.000 | .945      | .836     |
| TARGETPOP |       | 1.000     | .781     |
| DISPOINC  |       |           | 1.000    |

# Diagnostics and Remedial Measures, cont'd

- A complement to the scatter plot matrix that may be useful at times is the correlation matrix.
- This matrix contains the coefficients of simple correlation  $r_{Y,X1}$ ,  $r_{Y,X2}$ ,  $\dots$ ,  $r_{Y,Xp-1}$  between  $Y$  and each of the predictor variables.
- As well as all of the coefficients of simple correlation among the predictor variables  $r_{X1,X2}$ ,  $r_{X1,X3}$ , and etc.
- The format of the correlation matrix follows that of the scatter plot matrix:

$$\begin{bmatrix} 1 & r_{Y1} & r_{Y2} & \cdots & r_{Y,p-1} \\ r_{Y1} & 1 & r_{12} & \cdots & r_{1,p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ r_{Y,p-1} & r_{1,p-1} & r_{2,p-1} & \cdots & 1 \end{bmatrix}$$

# Diagnostics and Remedial Measures, cont'd

- Scatter Plots
- Residual Plots
- Correlation Test for Normality
- Test for Constancy of Error Variance
- F Test for Lack of Fit
- Box-Cox transformations

# Diagnostics and Remedial Measures, cont'd

All measures below discussed for simple linear regression can be carried over:

- Scatter Plots
- Residual Plots
- Correlation Test for Normality
- Test for Constancy of Error Variance
  - Brown-Forsythe Test
  - Breusch-Pagan Test
- F Test for Lack of Fit
- Box-Cox transformations

# F test of lack of fit

As discussed, to test whether the multiple regression response function

$$E\{Y\} = \beta_0 + \beta_1 X_1 + \cdots + \beta_{p-1} X_{p-1}$$

is an appropriate response surface.

- Repeat observations
- SSE is decomposed into Pure Error (PE) and Lack of Fit (LOF) components

# F test of lack of fit, cont'd

Testing:

$$H_0 : E\{Y\} = \beta_0 + \beta_1 X_1 + \cdots + \beta_{p-1} X_{p-1}$$

$$H_a : E\{Y\} \neq \beta_0 + \beta_1 X_1 + \cdots + \beta_{p-1} X_{p-1}$$

The test statistic:

$$F^* = \frac{SSLE}{c - p} \div \frac{SSPE}{n - c} = \frac{MSLF}{MSPE}$$

$\Rightarrow$  If  $F^* \leq F(1 - \alpha; c - p; n - c)$ , conclude  $H_0$

If  $F^* > F(1 - \alpha; c - p; n - c)$ , conclude  $H_a$

# Example: Dwaine Studio

```
> cor(DwaineStudio)
```

|  | X1 | X2 | Y |
|--|----|----|---|
|--|----|----|---|

|    |           |           |           |
|----|-----------|-----------|-----------|
| X1 | 1.0000000 | 0.7812993 | 0.9445543 |
|----|-----------|-----------|-----------|

|    |           |           |           |
|----|-----------|-----------|-----------|
| X2 | 0.7812993 | 1.0000000 | 0.8358025 |
|----|-----------|-----------|-----------|

|   |           |           |           |
|---|-----------|-----------|-----------|
| Y | 0.9445543 | 0.8358025 | 1.0000000 |
|---|-----------|-----------|-----------|

```
> anova(f)
```

Analysis of Variance Table

Response: Y

|           | Df | Sum Sq  | Mean Sq | F value  | Pr(>F)       |
|-----------|----|---------|---------|----------|--------------|
| X1        | 1  | 23371.8 | 23371.8 | 192.8962 | 4.64e-11 *** |
| X2        | 1  | 643.5   | 643.5   | 5.3108   | 0.03332 *    |
| Residuals | 18 | 2180.9  | 121.2   |          |              |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
>
```

```
> f<-lm(Y~X1+X2,data=DwaineStudio)
```

```
> summary(f)
```

Call:

lm(formula = Y ~ X1 + X2, data = DwaineStudio)

Residuals:

|  | Min      | 1Q      | Median | 3Q     | Max     |
|--|----------|---------|--------|--------|---------|
|  | -18.4239 | -6.2161 | 0.7449 | 9.4356 | 20.2151 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )  |
|-------------|----------|------------|---------|-----------|
| (Intercept) | -68.8571 | 60.0170    | -1.147  | 0.2663    |
| X1          | 1.4546   | 0.2118     | 6.868   | 2e-06 *** |
| X2          | 9.3655   | 4.0640     | 2.305   | 0.0333 *  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.01 on 18 degrees of freedom  
Multiple R-squared: 0.9167, Adjusted R-squared: 0.9075  
F-statistic: 99.1 on 2 and 18 DF, p-value: 1.921e-10